4 Theoretical guarantees

The first IRMF component fits a local constant to noisy observations with a Gaussian-smoothed absolute loss. This section bounds three errors of that fit. Section 4.1 defines the fit, two deterministic reference values, and the three errors. Section 4.2 bounds the *sampling error*, the random displacement of the fit from its population target. Section 4.3 bounds the *target shift*, the deterministic distance from the population target to the biased signal, which is the output of the same filter on noiseless data. Section 4.4 adds the two bounds to control the *biased signal error*. Section 4.5 records what the bounds do not say. All proofs are in Appendix A.

The conclusions are easy to summarize. Let n be the number of observations, h the window radius, and m the number of evaluation points. Under Gaussian noise and regular windows, the sampling error is of order $\sqrt{(x+\log(4m))/(nh)}$ uniformly over the evaluation points, with probability at least $1-e^{-x}$. The target shift is of order σ^2 for small noise level σ, and it does not decrease when n grows at a fixed filtering scale. As the sampling error vanishes, the biased signal error therefore tends to the target shift, not to zero.

4.1 Setting: three fits and three errors

Observations and weights. Let u range over n deterministic observation locations in $[0,1]$, and suppose that

$$Y_u = X_u + \varepsilon_u,$$

where X_u is the clean signal and the errors ε_u are independent and identically distributed with $\mathbb{E}|\varepsilon_u| < \infty$. Let $\mathcal{T}_n \subset [0,1]$ be a nonempty finite set of evaluation points and put $m = |\mathcal{T}_n|$. The signal, design, bandwidth, and weights are deterministic. For $t \in \mathcal{T}_n$, write

$$w_{u,t} := K_h(t-u), \qquad W_t := \sum_u w_{u,t}, \qquad Q_t^2 := \sum_u w_{u,t}^2, \qquad k_t := \max_u w_{u,t}, \tag{1}$$

and assume $w_{u,t} \geq 0$ and $W_t > 0$.

Loss. Let $\rho \in C^3(\mathbf{R})$ be convex and coercive, meaning $\rho(z) \to \infty$ as $|z| \to \infty$, with finite constants L_1, L_2, L_3 such that

$$|\rho'(z)| \leq L_1, \qquad 0 \leq \rho''(z) \leq L_2, \qquad |\rho'''(z)| \leq L_3, \qquad z \in \mathbf{R}. \tag{2}$$

The case of interest is the Gaussian-smoothed absolute value with a deterministic scale $H > 0$,

$$\begin{aligned} &\rho_H(r) := \mathbb{E}_Z|r + HZ|, \qquad Z \sim N(0,1), \\ &\rho_H'(r) = \operatorname{erf}\left(\frac{r}{\sqrt{2}H}\right), \qquad \rho_H''(r) = \frac{\sqrt{2/\pi}}{H} e^{-r^2/(2H^2)}, \qquad \rho_H'''(r) = -\frac{\sqrt{2/\pi}\, r}{H^3} e^{-r^2/(2H^2)}, \end{aligned} \tag{3}$$

where the auxiliary variable Z is independent of the observation noise. One may take the conservative constants

$$L_1 = 1, \qquad L_2 = \frac{2}{H\sqrt{2\pi}}, \qquad L_3 = \frac{2}{H^2\sqrt{2\pi}}. \tag{4}$$

Averaging the loss over the noise defines

$$\phi(z) := \mathbb{E}\rho(z+\varepsilon), \qquad \text{so that} \qquad \mathbb{E}\rho(Y_u - z) = \phi(X_u - z). \tag{5}$$

Independent errors with different distributions and finite first moments are also allowed, provided $\phi(X_u - z)$ is read as $\mathbb{E}\rho(Y_u - z)$ in each summand below.

Three fits. At each $t \in \mathcal{T}_n$, the same weights and the same loss define three minimizers:

$$\widetilde{S}_t := \operatorname*{arg\,inf}_{z\in\mathbf{R}} \sum_u \rho(Y_u - z)\, w_{u,t}, \qquad \text{observed component,}$$

$$S_t := \operatorname*{arg\,inf}_{z\in\mathbf{R}} \sum_u \phi(X_u - z)\, w_{u,t}, \qquad \text{population target,} \tag{6}$$

$$X_t^{\mathrm{b}} := \operatorname*{arg\,inf}_{z\in\mathbf{R}} \sum_u \rho(X_u - z)\, w_{u,t}, \qquad \text{biased signal.} \tag{7}$$

The observed component $\widetilde{S}_t$ is what the algorithm computes from the noisy data. The biased signal X_t^{b} is what the same algorithm would compute if the clean signal were available. The population target S_t minimizes the noisy loss after that loss has been averaged over noise realizations. Only $\widetilde{S}_t$ depends on the realized noise; for a fixed signal, design, and noise law, S_t and X_t^{b} are deterministic. Averaging a loss and then minimizing is generally different from averaging the minimizer, so S_t need not equal $\mathbb{E}\widetilde{S}_t$.

The name *biased signal* records that local filtering changes the signal even without observation noise: the difference $X_t^{\mathrm{b}} - X_t$ is deterministic filtering distortion. Nothing below assumes that the first component equals the signal value X_t, and nothing below bounds $X_t^{\mathrm{b}} - X_t$.

Existence and uniqueness. The Lipschitz bound $|\rho'| \le L_1$ and the finite first noise moment make the population objective finite, and the inequality $\mathbb{E}\rho(Y_u - z) \ge \rho(-z) - L_1\mathbb{E}|Y_u|$ makes it coercive. Thus all three minimizers exist. Define the population curvature and the centered empirical score at the population target by

$$F_t := \sum_u \phi''(X_u - S_t)\, w_{u,t}, \qquad D_t := F_t^{1/2}, \tag{8}$$

$$\nabla_t := \sum_u \big\{\rho'(Y_u - S_t) - \mathbb{E}\rho'(Y_u - S_t)\big\}\, w_{u,t}. \tag{9}$$

Require $F_t > 0$ for every $t \in \mathcal{T}_n$; convexity then makes S_t unique. Assume also that X_t^{b} is unique. For ρ_H, all three objectives are strictly convex, so all three minimizers are unique and $F_t > 0$ whenever $W_t > 0$. The objectives need not be strongly convex on $\mathbf{R}$, since their curvatures tend to zero in the tails. The argument uses positive curvature near S_t only.

Three errors. The three fits define three differences:

$\widetilde{S}_t - S_t$**: sampling error.** The displacement of the observed fit from its population target. This finite-sample error need not have mean zero.

$\Delta_t := S_t - X_t^{\mathrm{b}}$**: target shift.** The change in the minimizer caused by averaging the loss over observation noise. It can be nonzero even for mean-zero noise because the loss is nonlinear.

$\widetilde{S}_t - X_t^{\mathrm{b}}$**: biased signal error.** The total change in the extracted component caused by adding observation noise to the clean signal.

They satisfy the exact identity

$$\boxed{\widetilde{S}_t - X_t^{\mathrm{b}} = \underbrace{\widetilde{S}_t - S_t}_{\text{sampling error}} + \underbrace{\Delta_t}_{\text{target shift}}}. \tag{10}$$

Section 4.2 expands the sampling error as $\widetilde{S}_t - S_t = F_t^{-1}\nabla_t + R_t$ and bounds it, first for general noise and then explicitly for Gaussian noise. Section 4.3 bounds Δ_t, which contains no sampling fluctuation. Section 4.4 adds the two bounds on the same probability event.

4.2 Sampling error

4.2.1 Expansion for general noise

The sampling error is controlled by three quantities at the population target: the centered score ∇_t, the fluctuation of the empirical curvature

$$F_{\circ,t} := \sum_u \rho''(Y_u - S_t)\, w_{u,t}, \qquad \mathbb{E}F_{\circ,t} = F_t, \tag{11}$$

around F_t, and the size of the third derivative. The constants C_1, C_2, and C_3 bound these three quantities, in that order. The scale $D_t = \sqrt{F_t}$ expresses the first two in local curvature units. For $x \geq 0$ and $z_x := x + \log(4m)$, set

$$C_1 := \max_{t\in\mathcal{T}_n} \frac{L_1}{D_t}\left\{Q_t\sqrt{2z_x} + \frac{2k_t z_x}{3}\right\}, \tag{12}$$

$$C_2 := \max_{t\in\mathcal{T}_n} \frac{L_2}{D_t}\left\{Q_t\sqrt{2z_x} + \frac{k_t z_x}{3}\right\}, \tag{13}$$

$$C_3 := \max_{t\in\mathcal{T}_n} \frac{L_3 W_t}{D_t^2}, \qquad D_{\min} := \min_{t\in\mathcal{T}_n} D_t, \qquad \delta := \frac{C_2}{D_{\min}}. \tag{14}$$

The number x sets the failure probability e^{-x}, and the term $\log(4m)$ pays for a union bound over the evaluation grid. The parameter δ bounds the relative curvature error.

Lemma 4.1 (Score and curvature bounds)**.** *There is an event $\Omega(x)$ of probability at least $1 - e^{-x}$ on which, simultaneously for every $t \in \mathcal{T}_n$,*

(i) the centered score and the curvature fluctuation satisfy

$$\frac{|\nabla_t|}{D_t} \leq C_1, \qquad \frac{|F_{\circ,t} - F_t|}{D_t} \leq C_2; \tag{15}$$

(ii) the empirical third derivative satisfies, deterministically,

$$\sup_{z\in\mathbf{R}} \frac{1}{D_t^2} \left| \sum_u \rho'''(Y_u - z)\, w_{u,t} \right| \le C_3; \tag{16}$$

(iii) the population curvature satisfies, deterministically, for every $r > 0$ and every z with $D_t|z - S_t| \le r$,

$$\frac{1}{D_t^2} \left| \sum_u \left\{ \phi''(X_u - z) - \phi''(X_u - S_t) \right\} w_{u,t} \right| \le \frac{C_3 r}{D_t}. \tag{17}$$

Part (i) uses Bernstein's inequality and a union bound over the finite grid; parts (ii) and (iii) follow from (2). The proposition below uses only these bounds, so it also applies with any other constants that satisfy them on a specified event.

Write the sampling error as a linear term plus a remainder,

$$\widetilde{S}_t - S_t = F_t^{-1}\nabla_t + R_t, \qquad \widetilde{S}_t^{\mathrm{corr}} := \widetilde{S}_t - F_t^{-1}\nabla_t = S_t + R_t. \tag{18}$$

The oracle-corrected estimator $\widetilde{S}_t^{\mathrm{corr}}$ subtracts the leading sampling fluctuation. It is an expansion tool, not a computable estimator: it uses the unknown target, noise law, and curvature.

Proposition 4.2 (Sampling error expansion)**.** *Suppose that the conditions of Section 4.1 hold and that*

$$\delta < 1, \qquad \frac{C_1 C_3}{D_{\min}(1-\delta)^2} < \frac{4}{9}. \tag{19}$$

Then on the event $\Omega(x)$ of Lemma 4.1, every empirical minimizer $\widetilde{S}_t$ is unique, and for every $t \in \mathcal{T}_n$

$$|\widetilde{S}_t - S_t| \le \frac{3C_1}{2D_t(1-\delta)}, \tag{20}$$

$$D_t\,|R_t| \le \frac{1}{D_{\min}} \left\{ \frac{9C_1^2 C_3}{8(1-\delta)^3} + \frac{C_1 C_2}{1-\delta} \right\}. \tag{21}$$

The two conditions in (19) keep the empirical curvature positive near S_t and make a scalar Taylor expansion valid at the empirical minimizer. The first bound is a localization of $\widetilde{S}_t$ around S_t; the second shows that the remainder is of second order in C_1.

Remark (Sign of the correction)**.** Since the population score vanishes at S_t, the derivative of the empirical objective at S_t equals $-\nabla_t$; see (53). Thus $\widetilde{S}_t - S_t$ has first-order term $+F_t^{-1}\nabla_t$, and one subtracts that term to correct $\widetilde{S}_t$. This removes the leading sampling fluctuation; it does not touch Δ_t.

4.2.2 Explicit constants for Gaussian noise

Multiplying all weights in a window by a positive constant changes neither the three minimizers nor $F_t^{-1}\nabla_t$. This subsection therefore uses normalized weights,

$$a_{u,t} := \frac{w_{u,t}}{W_t}, \qquad N := \min_t \frac{1}{\sum_u a_{u,t}^2}, \qquad \alpha := \max_{t,u} a_{u,t}, \tag{22}$$

where N is the smallest effective sample size across windows and α the largest single weight. The normalized population curvature and score are $\bar{F}_t = F_t/W_t$ and $\bar{\nabla}_t = \nabla_t/W_t$; their ratio is still $F_t^{-1}\nabla_t$.

Assume now $\varepsilon_u \sim N(0, \sigma^2)$ independently, with $\sigma \geq 0$, and $\rho = \rho_H$. Put

$$c := \sqrt{2/\pi}, \qquad B := \sqrt{H^2 + \sigma^2}. \tag{23}$$

Since $\varepsilon + HZ \sim N(0, B^2)$, the averaged loss is again a smoothed absolute value, at the larger scale B:

$$\phi(r) = \mathbb{E}_\varepsilon \rho_H(r + \varepsilon) = \rho_B(r). \tag{24}$$

Thus X_t^{b} fits the clean signal at scale H, whereas S_t fits it at scale B. Explicitly, the two targets solve

$$\sum_u a_{u,t} \operatorname{erf}\left(\frac{X_u - X_t^{\mathrm{b}}}{\sqrt{2}H}\right) = 0, \qquad \sum_u a_{u,t} \operatorname{erf}\left(\frac{X_u - S_t}{\sqrt{2}B}\right) = 0, \tag{25}$$

and the normalized curvature at the target is

$$\bar{F}_t = \frac{c}{B} \sum_u a_{u,t} e^{-(X_u - S_t)^2/(2B^2)}. \tag{26}$$

Each weighted score in (25) is strictly decreasing in the fitted value, positive below all active signal values and negative above them. Both X_t^{b} and S_t therefore lie in the *active signal range*, the interval between the smallest and largest X_u with $a_{u,t} > 0$. Let

$$A_t := \max_{u: a_{u,t} > 0} X_u - \min_{u: a_{u,t} > 0} X_u, \qquad \kappa_t := \frac{c}{B} e^{-A_t^2/(2B^2)} \tag{27}$$

be the width of this range and the resulting curvature bound. For every z in the active range and every active u, $|X_u - z| \leq A_t$, so the normalized population curvature at scale B is at least κ_t throughout the range; in particular $\bar{F}_t \geq \kappa_t$. Fix a curvature lower bound

$$0 < \lambda \leq \min_t \bar{F}_t, \qquad \text{for instance} \qquad \lambda = \min_t \kappa_t. \tag{28}$$

Any known positive lower bound may be used; a cruder choice is $\lambda = \frac{c}{B} e^{-A^2/(2B^2)}$ with $A \geq \max_t A_t$.

The Gaussian model permits variance bounds sharper than the crude L_1^2 and L_2^2: the score and curvature summands are Lipschitz in the noise, so their variances are at most $\sigma^2 L_2^2$ and $\sigma^2 L_3^2$. With $L_2 = c/H$ and $L_3 = c/H^2$, define

$$\begin{aligned} v_1 &:= \min\{1, \sigma L_2\}, & v_2 &:= \min\{L_2, \sigma L_3\}, \\ s_1 &:= v_1 \sqrt{\frac{2z_x}{N}} + \frac{2\alpha z_x}{3}, & s_2 &:= v_2 \sqrt{\frac{2z_x}{N}} + \frac{L_2 \alpha z_x}{3}, \\ d &:= \frac{s_2}{\lambda}, & z_x &= x + \log(4m). \end{aligned} \tag{29}$$

Here s_1 and s_2 bound the normalized score and curvature fluctuations, and d is the relative curvature perturbation. They play the roles of C_1, C_2, and δ: the proof applies Proposition 4.2 to the normalized objectives with $C_1 = s_1/\sqrt{\lambda}$, $C_2 = s_2/\sqrt{\lambda}$, $C_3 = L_3/\lambda$, $D_{\min} \geq \sqrt{\lambda}$, and $\delta \leq d$.

Corollary 4.3 (Explicit Gaussian bounds)**.** *Under the Gaussian model and the deterministic design conditions above, suppose*

$$d < 1, \qquad \frac{s_1 L_3}{\lambda^2 (1-d)^2} < \frac{4}{9}. \tag{30}$$

With probability at least $1 - e^{-x}$*, simultaneously for all* t*,*

$$|\widetilde{S}_t - S_t| \le \mathcal{E}_x := \frac{3 s_1}{2\lambda(1-d)}, \tag{31}$$

$$|F_t^{-1} \nabla_t| \le \frac{s_1}{\lambda}, \tag{32}$$

$$|R_t| \le \mathcal{R}_x := \frac{9 L_3 s_1^2}{8\lambda^3 (1-d)^3} + \frac{s_1 s_2}{\lambda^2 (1-d)}, \tag{33}$$

and in particular

$$|\widetilde{S}_t^{\mathrm{corr}} - S_t| \le \mathcal{R}_x. \tag{34}$$

The constants $\mathcal{E}_x$ and $\mathcal{R}_x$ are determined by the weights, x, H, σ, and the curvature lower bound λ. They bound the sampling error only; the target shift enters in Section 4.4.

Remark (Dependence on the noise and loss scales)**.** The noise scale enters through $B = \sqrt{H^2 + \sigma^2}$, the lower bound λ, and the variance bounds v_1, v_2. These are sufficient, conservative constants. At $\sigma = 0$, the empirical and population objectives coincide with the clean objective, so both errors and R_t vanish; the Bernstein linear terms need not vanish. If H varies with sample size, the explicit conditions must be checked again: L_2 and L_3 grow as H^{-1} and H^{-2} when H decreases.

4.2.3 Rates in the local sample size

The bounds of Corollary 4.3 become rates once the effective sample size is related to nh. Suppose that, for a known constant $C_K > 0$,

$$N \ge \frac{nh}{C_K}, \qquad \alpha \le \frac{C_K}{nh}. \tag{35}$$

These assumptions say that the effective sample size is at least a constant times nh, and that no individual observation has disproportionate weight. They can be checked directly from the normalized weights.

They hold for equally spaced designs. Let Δ be the grid spacing, take $0 < h < 1/2$, and let $w_{u,t} = K((t-u)/h)$ with K supported on $[-1,1]$, $0 \le K \le K_{\max}$, and $K(v) \ge k_0 > 0$ on $|v| \le \eta$ for some $0 < \eta < 1$. Use untruncated interior windows, or a periodic grid with local offsets in which each of the n locations is counted once. If $\eta h \ge \Delta$, the central interval $|t-u| \le \eta h$ contains at least $\lfloor 2\eta h/\Delta \rfloor \ge \eta h/\Delta$ grid points, so

$$W_t \ge \frac{k_0 \eta h}{\Delta}, \qquad \max_u a_{u,t} \le \frac{K_{\max}\Delta}{k_0 \eta h}, \qquad \sum_u a_{u,t}^2 \le \max_u a_{u,t}.$$

Unit-interval grids have $\Delta = 1/n$ when the right endpoint is excluded and $\Delta = 1/(n-1)$ when both endpoints are included; for $n \ge 2$, both satisfy $\Delta \le 2/n$. Hence the single choice

$$C_K = \frac{2 K_{\max}}{k_0 \eta}$$

establishes (35) for either convention whenever $\eta h \geq \Delta$, for which $nh \geq 2/\eta$ suffices. If the spacing is exactly $1/n$, the sharper constant $C_K = K_{\max}/(k_0\eta)$ and the condition $nh \geq 1/\eta$ suffice. Conversely, at most $2h/\Delta + 1$ observations are active in any window, so the effective sample size is also at most of order nh.

Set

$$q_n := \sqrt{\frac{C_K z_x}{nh}}, \qquad A_1 := \sqrt{2}v_1 + \frac{2}{3}, \qquad A_2 := \sqrt{2}v_2 + \frac{L_2}{3}. \tag{36}$$

Corollary 4.4 (A sufficient local sample size)**.** *Under the Gaussian model, the curvature lower bound* (28)*, and* (35)*, the condition*

$$nh > C_K z_x \max\left\{1, \frac{4A_2^2}{\lambda^2}, \frac{81A_1^2L_3^2}{\lambda^4}\right\} \tag{37}$$

implies (30)*, and with probability at least* $1 - e^{-x}$*,*

$$\max_t |\widetilde{S}_t - S_t| \leq \frac{3A_1}{\lambda} q_n, \tag{38}$$

$$\max_t |R_t| \leq \left\{\frac{9L_3A_1^2}{\lambda^3} + \frac{2A_1A_2}{\lambda^2}\right\} q_n^2. \tag{39}$$

For fixed $H > 0$, fixed σ, bounded local signal oscillation, and fixed kernel constants, λ stays bounded away from zero. The two bounds then have orders $\sqrt{z_x/(nh)}$ and $z_x/(nh)$, where $z_x = x + \log(4m)$, and the smallness conditions hold eventually if $z_x/(nh) \to 0$. These are upper bounds on the sampling error; they do not assert that the remainder equals the squared realized error.

4.3 Target shift

The bounds of Section 4.2 are relative to S_t. By (25), the biased signal X_t^{b} is the zero of the clean score at scale H, while S_t is the zero of the same score at scale $B > H$ whenever $\sigma > 0$. The change of scale moves the zero, and the displacement $\Delta_t = S_t - X_t^{\mathrm{b}}$ does not depend on n. This subsection bounds it.

Recall from (27) the width A_t of the active signal range and the curvature bound κ_t. Both S_t and X_t^{b} lie in the active range, and throughout the interval between them the normalized population curvature at scale B is at least κ_t. This interval bound is what the mean value argument needs; a curvature bound at S_t alone would not suffice.

Proposition 4.5 (Deterministic target shift)**.** *Under the Gaussian model, put*

$$\beta_t := \min\left\{A_t, \frac{L_3\sigma^2}{2\kappa_t}\right\}, \qquad \beta := \max_t \beta_t. \tag{40}$$

Then, deterministically,

$$|\Delta_t| = |S_t - X_t^{\mathrm{b}}| \leq \beta_t \qquad \textit{for every } t. \tag{41}$$

For fixed signal, weights, and H, the shift bound is $O(\sigma^2)$ as $\sigma \to 0$. It is a sufficient upper bound and can be conservative. At $\sigma = 0$ the shift vanishes exactly. It also vanishes when the weighted signal values are symmetric about a number μ_t: both scores in (25) are then zero at μ_t, so $X_t^{\mathrm{b}} = S_t = \mu_t$ even when $\sigma > 0$. This includes constant signals. Symmetric observation locations alone do not imply symmetry of signal values for a curved signal.

Example (An asymmetric window)**.** Take equally weighted signal values $(0, 0, 3)$, $H = 0.5$, and $\sigma = 1$. The target at scale b is the unique root on $[0, 3]$ of

$$2\,\mathrm{erf}\left(\frac{-z}{\sqrt{2}b}\right) + \mathrm{erf}\left(\frac{3-z}{\sqrt{2}b}\right) = 0.$$

Bisection at $b = H$ and $b = B = \sqrt{1.25}$ gives

$$X_t^{\mathrm{b}} \approx 0.337245, \qquad S_t \approx 0.718197, \qquad \Delta_t \approx 0.380952.$$

Repeating each signal value r times, with weights $1/(3r)$ and independent observation noise, preserves both targets and gives effective sample size $3r$. As $r \to \infty$, the sampling error tends to zero in probability, while the biased signal error tends to Δ_t.

4.4 Biased signal error

The sampling error and the target shift now have separate bounds, the first on a probability event and the second deterministic. Substituting the expansion (18) into the identity (10) gives

$$\widetilde{S}_t - X_t^{\mathrm{b}} = \underbrace{F_t^{-1}\nabla_t + R_t}_{\text{sampling error}} + \underbrace{\Delta_t}_{\text{target shift}}, \qquad \widetilde{S}_t^{\mathrm{corr}} - X_t^{\mathrm{b}} = R_t + \Delta_t. \tag{42}$$

The oracle correction removes the leading sampling term and leaves the shift untouched.

Corollary 4.6 (Combined error bounds)**.** *Under the Gaussian model, if the smallness conditions* (30) *hold, then with probability at least* $1 - e^{-x}$*, simultaneously for all* t*,*

$$|\widetilde{S}_t - S_t| \le \mathcal{E}_x, \tag{43}$$

$$|\widetilde{S}_t - X_t^{\mathrm{b}}| \le \beta_t + \mathcal{E}_x, \tag{44}$$

$$|\widetilde{S}_t^{\mathrm{corr}} - S_t| \le \mathcal{R}_x, \tag{45}$$

$$|\widetilde{S}_t^{\mathrm{corr}} - X_t^{\mathrm{b}}| \le \beta_t + \mathcal{R}_x. \tag{46}$$

Under the additional conditions of Corollary 4.4, also

$$\max_t |\widetilde{S}_t - X_t^{\mathrm{b}}| \le \beta + \frac{3A_1}{\lambda} q_n, \tag{47}$$

$$\max_t |\widetilde{S}_t^{\mathrm{corr}} - X_t^{\mathrm{b}}| \le \beta + \left\{ \frac{9L_3 A_1^2}{\lambda^3} + \frac{2A_1 A_2}{\lambda^2} \right\} q_n^2. \tag{48}$$

Replacing β_t by β in (44) and (46) gives uniform bounds over the evaluation grid. The shift bound is deterministic, so adding it does not change the failure probability. The sampling contribution decreases at the stated rates, while the target shift remains.

The shift is a floor, not only an upper bound. For any realization, the reverse triangle inequality applied to (10) gives

$$\left| \max_t |\widetilde{S}_t - X_t^{\mathrm{b}}| - \max_t |\Delta_t| \right| \le \max_t |\widetilde{S}_t - S_t|. \tag{49}$$

Thus a nonzero shift Δ_t persists as the sampling error decreases. The upper bound β itself is not a lower bound on the error.

4.5 Scope of the conclusions

All uniform probability statements concern the finite evaluation grid. A supremum over the continuum $[0,1]$ requires additional regularity and control of the family of weighted processes. Adaptive choices of bandwidth or loss scale also require simultaneous control over those choices.

The oracle correction uses the unknown population target, noise law, and curvature. It is an expansion tool, not an estimator computable from the observations alone. Its remainder bound around S_t becomes a bound around X_t^{b} only after adding the target shift. Neither signed error is guaranteed to have the smaller absolute value on every realization.

In synthetic experiments, compute X^{b} by applying the filter to the clean signal at scale H, and compute S at scale B under Gaussian noise. Evaluate both errors on the same noise realization and report $S - X^{\mathrm{b}}$. Comparing with S tests the stochastic expansion; comparing with X^{b} measures how noise changes the biased signal. Neither comparison alone establishes recovery of latent generative components.

Proposition 4.2 allows independent non-Gaussian errors with a finite first moment. The explicit Gaussian formulas for the population target and its shift do not cover arbitrary non-Gaussian noise. Later IRMF components use estimated residuals that share observations, so their dependent errors require a separate argument.

Throughout, F_t is evaluated at S_t. It cannot be replaced by a curvature evaluated at X_t^{b} without bounding their difference.

A Proofs

The argument has two stages. Lemma 4.1 controls the random score and curvature at the population target. Conditional on that event, the proof of Proposition 4.2 is deterministic: localize the empirical minimizer, apply a scalar Taylor expansion, and replace empirical curvature by population curvature. The remaining proofs specialize these bounds and control the target shift.

Notation for the proofs. Fix $t \in \mathcal{T}_n$; all quantities below may depend on t. Abbreviate the population and empirical objectives as

$$f(z) := \sum_u \phi(X_u - z) w_{u,t}, \qquad f_\circ(z) := \sum_u \rho(Y_u - z) w_{u,t}, \tag{50}$$

their minimizers as

$$x^* := \arg\inf_z f(z) = S_t, \qquad x_\circ := \arg\inf_z f_\circ(z) = \widetilde{S}_t, \tag{51}$$

and their curvatures at the population target as

$$F_\circ := f_\circ''(x^*) = F_{\circ,t}, \qquad F := f''(x^*) = F_t = D^2. \tag{52}$$

Differentiating (50) gives

$$f'(z) = -\sum_u \phi'(X_u - z) w_{u,t}, \qquad f_\circ'(z) = -\sum_u \rho'(Y_u - z) w_{u,t},$$
$$f''(z) = \sum_u \phi''(X_u - z) w_{u,t}, \qquad f_\circ''(z) = \sum_u \rho''(Y_u - z) w_{u,t}.$$

Differentiation under the expectation is justified by (2), so $\phi^{(j)}(a) = \mathbb{E}\rho^{(j)}(a+\varepsilon)$ for $j = 1, 2, 3$. Since $f'(x^*) = 0$,

$$f'_\circ(x^*) = -\sum_u \left\{\rho'(Y_u - x^*) - \mathbb{E}\rho'(Y_u - x^*)\right\} w_{u,t} = -\nabla_t, \tag{53}$$

$$F_\circ - F = \sum_u \left\{\rho''(Y_u - x^*) - \mathbb{E}\rho''(Y_u - x^*)\right\} w_{u,t}. \tag{54}$$

For $r > 0$, let $U_t(r) := \{z \in \mathbf{R} : |D(z - x^*)| \le r\}$.

A.1 Score and curvature concentration

Proof of Lemma 4.1. For the score bound in part (i), set

$$Z_{u,t} := w_{u,t}\left\{\rho'(Y_u - x^*) - \mathbb{E}\rho'(Y_u - x^*)\right\},$$

so that $\nabla_t = \sum_u Z_{u,t}$ by (53). These variables are independent and centered. Moreover,

$$|Z_{u,t}| \le 2L_1 k_t, \qquad \sum_u \mathbb{E}Z_{u,t}^2 \le L_1^2 Q_t^2.$$

Bernstein's inequality therefore yields, for every $z > 0$,

$$\mathbf{P}\left(\left|\sum_u Z_{u,t}\right| > L_1\left\{Q_t\sqrt{2z} + \frac{2k_t z}{3}\right\}\right) \le 2e^{-z}. \tag{55}$$

For the curvature bound, use

$$\widetilde{Z}_{u,t} := w_{u,t}\left\{\rho''(Y_u - x^*) - \mathbb{E}\rho''(Y_u - x^*)\right\}, \qquad F_\circ - F = \sum_u \widetilde{Z}_{u,t}.$$

Convexity and (2) give

$$|\widetilde{Z}_{u,t}| \le L_2 k_t, \qquad \sum_u \mathbb{E}\widetilde{Z}_{u,t}^2 \le L_2^2 Q_t^2.$$

Hence

$$\mathbf{P}\left(\left|\sum_u \widetilde{Z}_{u,t}\right| > L_2\left\{Q_t\sqrt{2z} + \frac{k_t z}{3}\right\}\right) \le 2e^{-z}. \tag{56}$$

Take $z = z_x = x + \log(4m)$ in (55) and (56), and take the union over the two inequalities and over $t \in \mathcal{T}_n$. The exceptional probability is at most $4me^{-z_x} = e^{-x}$. Dividing by D_t and using (12) and (13) proves part (i).

Part (ii) is deterministic. By (2) and (14),

$$\sup_{z\in\mathbf{R}} \frac{1}{D^2}\left|\sum_u \rho'''(Y_u - z) w_{u,t}\right| \le \frac{L_3 W_t}{D^2} \le C_3.$$

Finally, $|\phi'''| \le L_3$. The mean value theorem and $|z - x^*| \le r/D$ for $z \in U_t(r)$ give

$$\frac{1}{D^2}\left|\sum_u \left\{\phi''(X_u - z) - \phi''(X_u - x^*)\right\} w_{u,t}\right| \le \frac{L_3 W_t}{D^2}|z - x^*| \le \frac{C_3 r}{D},$$

which proves part (iii). □

A.2 Scalar localization and the remainder

Proof of Proposition 4.2. Fix $t \in \mathcal{T}_n$ and work on the event of Lemma 4.1. By (53) and (15), $|f_\circ'(x^*)| = |\nabla_t| \le C_1 D$ and

$$\frac{|F_\circ - F|}{F} \le \frac{C_2}{D} \le \delta. \tag{57}$$

In particular, $F_\circ \ge (1-\delta)F > 0$. With

$$b_\circ := \frac{D}{F_\circ}|f_\circ'(x^*)|, \tag{58}$$

the score bound and (57) imply

$$b_\circ \le \frac{C_1}{1-\delta}. \tag{59}$$

Use the scaled coordinate $v = D(z - x^*)$, and put

$$g(v) := f_\circ(x^* + v/D), \qquad a := g''(0) = F_\circ/D^2, \qquad \tau := C_3/D. \tag{60}$$

Then $a \ge 1-\delta$, $|g'''(v)| \le \tau$ by (16), and $b_\circ = |g'(0)|/a$. Consequently,

$$\frac{\tau b_\circ}{a} \le \frac{C_1 C_3}{D(1-\delta)^2} < \frac{4}{9}. \tag{61}$$

If $b_\circ = 0$, convexity and $g''(0) > 0$ imply that 0 is the unique minimizer, and the claimed bounds are immediate. Otherwise set $r = 3b_\circ/2$. Taylor expansion of g' at zero yields

$$|g'(v) - g'(0) - av| \le \frac{\tau}{2}v^2.$$

Since $|g'(0)| = ab_\circ$,

$$g'(r) \ge -ab_\circ + ar - \frac{\tau r^2}{2} = \frac{ab_\circ}{2} - \frac{9\tau b_\circ^2}{8} > 0,$$
$$g'(-r) \le ab_\circ - ar + \frac{\tau r^2}{2} = -\frac{ab_\circ}{2} + \frac{9\tau b_\circ^2}{8} < 0.$$

Moreover, for $|v| \le r$,

$$g''(v) \ge a - \tau r > \frac{a}{3} > 0.$$

There is therefore a unique zero $v_\circ \in (-r, r)$ of g'. Convexity on $\mathbf{R}$ makes it the unique global minimizer: the derivative is negative to the left of this interval and positive to its right. This proves uniqueness of $x_\circ$ and

$$|v_\circ| = D|x_\circ - x^*| \le \frac{3}{2}b_\circ \le \frac{3C_1}{2(1-\delta)}, \tag{62}$$

which is (20). At the zero of g', the same Taylor expansion gives

$$\left| a\left(v_\circ + \frac{g'(0)}{a}\right)\right| \le \frac{\tau}{2}v_\circ^2 \le \frac{9\tau}{8}b_\circ^2. \tag{63}$$

Equivalently, the left-hand side is $D^{-1}F_\circ|x_\circ - x^* + F_\circ^{-1}f_\circ'(x^*)|$. Using (57) and (59), we obtain

$$\left|x_\circ - x^* + F_\circ^{-1}f_\circ'(x^*)\right| \le \frac{9C_1^2 C_3}{8D^2(1-\delta)^3}. \tag{64}$$

It remains to replace $F_\circ$ by the population curvature F. By (57) and $|f'_\circ(x^*)| \le C_1 D$,

$$\begin{aligned}\left|F_\circ^{-1} f'_\circ(x^*) - F^{-1} f'_\circ(x^*)\right| &= \frac{|F_\circ - F|}{F_\circ F}|f'_\circ(x^*)| \\ &\le \frac{C_1 C_2}{D^2(1-\delta)}. \end{aligned} \tag{65}$$

The triangle inequality applied to (64) and (65), followed by $f'_\circ(x^*) = -\nabla_t$, proves

$$D\left|x_\circ - x^* - F^{-1}\nabla_t\right| \le \frac{1}{D}\left\{\frac{9C_1^2 C_3}{8(1-\delta)^3} + \frac{C_1 C_2}{1-\delta}\right\}. \tag{66}$$

The left-hand side is $D_t|R_t|$, and $D_t \ge D_{\min}$ gives (21). □

Remark (The remainder coefficient)**.** The coefficient $9/8$ is $\frac{1}{2}(\frac{3}{2})^2$. The factor $1/2$ comes from Taylor's theorem and $3/2$ from the localization radius.

A.3 Gaussian constants

Proof of Corollary 4.3. The identity (24) follows because $\varepsilon + HZ \sim N(0, B^2)$; the target and curvature formulas (25) and (26) follow by differentiation. For a Lipschitz function G and an independent copy ε' of ε,

$$\mathrm{Var}\{G(X_u + \varepsilon)\} = \tfrac{1}{2}\mathbb{E}\{G(X_u + \varepsilon) - G(X_u + \varepsilon')\}^2 \le \mathrm{Lip}(G)^2\sigma^2.$$

Apply this with $G(y) = \rho'_H(y - S_t)$ and $G(y) = \rho''_H(y - S_t)$. Their Lipschitz constants are at most L_2 and L_3. Combining this estimate with the boundedness estimates in the proof of Lemma 4.1 gives variance bounds v_1^2 and v_2^2 per unit squared weight. Bernstein's inequality and the same union bound therefore give an event of probability at least $1 - e^{-x}$ on which

$$\max_t |\bar\nabla_t| \le s_1, \qquad \max_t |\bar F_{\circ,t} - \bar F_t| \le s_2, \qquad \bar F_{\circ,t} = F_{\circ,t}/W_t = \sum_u a_{u,t}\rho''_H(Y_u - S_t).$$

For the normalized objectives, the deterministic proof of Proposition 4.2 applies with the bounds

$$C_1 = \frac{s_1}{\sqrt{\lambda}}, \qquad C_2 = \frac{s_2}{\sqrt{\lambda}}, \qquad C_3 = \frac{L_3}{\lambda}, \qquad D_{\min} \ge \sqrt{\lambda}, \qquad \delta \le d.$$

Condition (30) is sufficient for (19). Substitution in (20), using $D_t \ge \sqrt{\lambda}$, proves the raw bound. Substitution in (66), followed by division by $D_t \ge \sqrt{\lambda}$, proves the remainder bound. The score bound and $\bar F_t \ge \lambda$ give (32). The identity $\widetilde S_t^{\mathrm{corr}} - S_t = R_t$ gives (34). □

A.4 Sufficient local sample size

Proof of Corollary 4.4. The mass bounds imply

$$s_1 \le \sqrt{2}v_1 q_n + \tfrac{2}{3}q_n^2, \qquad s_2 \le \sqrt{2}v_2 q_n + \tfrac{L_2}{3}q_n^2.$$

Condition (37) implies

$$q_n < \min\left\{1, \frac{\lambda}{2A_2}, \frac{\lambda^2}{9A_1 L_3}\right\}.$$

Therefore $s_1 \le A_1 q_n$, $s_2 \le A_2 q_n$, $d < 1/2$, and

$$\frac{s_1 L_3}{\lambda^2(1-d)^2} \le \frac{4A_1 q_n L_3}{\lambda^2} < \frac{4}{9}.$$

Apply Corollary 4.3 and use $(1-d)^{-1} \le 2$ and $(1-d)^{-3} \le 8$. □

A.5 Target shift and biased signal error

Proof of Proposition 4.5. Write

$$\psi_b(r) = \operatorname{erf}\left(\frac{r}{\sqrt{2}b}\right), \qquad G_{b,t}(z) = \sum_u a_{u,t}\psi_b(X_u - z).$$

Both scores $G_{H,t}$ and $G_{B,t}$ are strictly decreasing, positive below the active signal range, and negative above it. Their zeros X_t^{b} and S_t therefore lie in that range. In particular, $|S_t - X_t^{\mathrm{b}}| \le A_t$.

Differentiating the Gaussian convolution identity gives $\psi_B(r) = \mathbb{E}\psi_H(r+\varepsilon)$. Since $\sup_r |\psi_H''(r)| = \sup_r |\rho_H'''(r)| \le L_3$, Taylor's theorem gives

$$|\psi_H(r+e) - \psi_H(r) - e\psi_H'(r)| \le \frac{L_3}{2}e^2 \qquad (r, e \in \mathbf{R}).$$

Taking expectation with $e = \varepsilon$, and using $\mathbb{E}\varepsilon = 0$ and $\mathbb{E}\varepsilon^2 = \sigma^2$, yields

$$|\psi_B(r) - \psi_H(r)| \le \frac{L_3\sigma^2}{2}.$$

As $G_{H,t}(X_t^{\mathrm{b}}) = 0$ and the weights sum to one,

$$|G_{B,t}(X_t^{\mathrm{b}})| = \left|\sum_u a_{u,t}\{\psi_B(X_u - X_t^{\mathrm{b}}) - \psi_H(X_u - X_t^{\mathrm{b}})\}\right| \le \frac{L_3\sigma^2}{2}. \tag{67}$$

For every z between X_t^{b} and S_t, active residuals satisfy $|X_u - z| \le A_t$. Hence

$$-G_{B,t}'(z) = \frac{c}{B}\sum_u a_{u,t}e^{-(X_u - z)^2/(2B^2)} \ge \kappa_t. \tag{68}$$

Since $G_{B,t}(S_t) = 0$, the mean value theorem gives

$$|S_t - X_t^{\mathrm{b}}| \le \frac{|G_{B,t}(X_t^{\mathrm{b}})|}{\kappa_t} \le \frac{L_3\sigma^2}{2\kappa_t}.$$

Combining this with the range bound proves (41). Evaluating (68) at $z = S_t$ also confirms $\bar{F}_t \ge \kappa_t$, as claimed in Section 4.2.2. □

Proof of Corollary 4.6. On the event of Corollary 4.3, add the deterministic bound $|\Delta_t| \le \beta_t$ to (31) and to (34). Identity (42) and the triangle inequality give (43)–(46) without changing the exceptional probability. Adding β to the bounds in Corollary 4.4 gives (47) and (48). □